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AR HUD Interface Optimization Model for Balancing Driver's Visual Sensitivity and Fatigue

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Abstract

With the wide application of intelligent driving assistance system, the augmented reality head-up display (AR HUD) interface plays an increasingly important role in drivers' real-time perception of traffic environment. As the AR HUD is vulnerable to be affected by changed ambient light, it weakens the legibility of the virtual interface and driver's visual comfort to a certain extent. The content color, as one of interface visual design elements, directly affects the performance of driver, which include the visual sensitivity and fatigue and further influences the vehicle safety. The selection between luminance contrast (affects sensitive and number of applicable environments) and driver performance (affects fatigue) are contradictory issues. In order to solve this contradictory problem in the design of AR HUD interface, and balance the cognitive load and situation awareness of driver, an extended analysis and optimization method of interface element was proposed. This method quantifies the discrete interface elements and makes the opposition problem consistent. The content color selection models under two conditions are constructed, which describe the distance from the chosen color to the optimal interval. It has been proved to be effective in finding the best answer for driver performance in interface design. Other interface elements in AR HUD can also be further optimized based on this method.

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1. Introduction

The AR HUD could enhance driving-related information (such as speed, warning signals, navigation arrows) to the windshield of the vehicle to maximize the use of visual resources and allow the driver to focus on safe driving (See Fig.1). It represents the most innovative development direction of intelligent vehicles and becomes an important HMI in the era of the Internet of Things. With the wide application of intelligent driving assistance system, the demand of AR HUD is rising rapidly, and its market space has a fast expansion. As the main connection points between the driver and the vehicle, these in-vehicle HMI demand great care and attention from manufacturers, designers and human factor engineers [1].

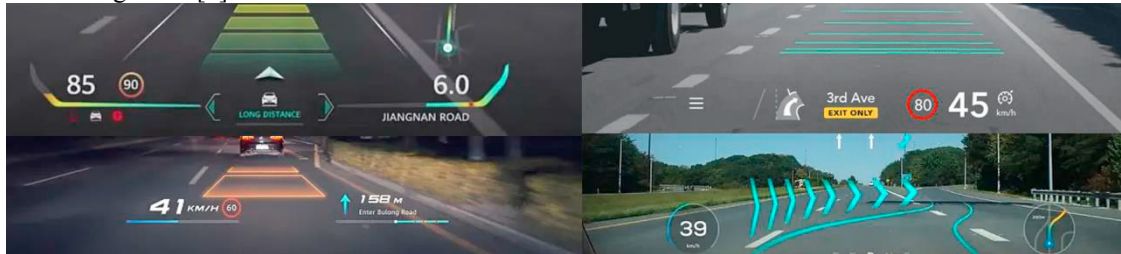


Fig. 1. AR HUD interface.

The AR HUD interface is an essential way to exchange information between drivers and automatic systems. It is also the terminal for information collection and output in intelligent vehicles. Consequently, the drivers execute and complete a different kind of tasks, such as perceive surrounding situation, navigation guidance [2], road recognition [3], and decision-making analysis [4] based on the real time graphical and text information displayed in AR HUD interface. But unlike the traditional displays, the virtual contents on the AR HUD are superimposed displayed optically superimposed on the real word though special optical system, which causes it to be transparent. As a result, the changeable environmental context and complex ambient light weaken the legibility of AR contents. And it also reduced the visual comfort and cognitive performance of drivers [5].

Based on the design principles of human-vehicle interaction system, AR HUD interface needs to meet drivers' recognition and reading of visual information efficiently, and reduce drivers' distraction and cognitive load during driving as well [6-7]. The AR system which poorly designed or ignored the principle of human factors is likely to fail to achieve its goal of helping human beings, and may even seriously damage human performance. As an ergonomics factor, content color directly affects driver performance, including visual sensitivity and fatigue. On the one hand, the dark color, such as violet, can achieves high luminance contrast, which means driver's information sensitivity improved, but just perform well in a short time. On the other hand, the light color, such as yellow, has low luminance contrast for driver while gazing at AR HUD. They achieve better performance with low visual fatigue during a long time, but less sensitivity. Therefore, this paper is devoted to find the content color value that can comprehensively consider the driver's visual sensitivity and fatigue, and quantify the selection process. This study also has reference significance for balancing situation awareness and cognitive load of drivers.

The paper is organized as follows. First, the main theories and methods which supporting the interface optimization model are introduced in Section 2. Then, the content color and its extended matter-elements of AR HUD interface are modeled and analysis in the Section 3. Section 4 describes two mathematical color selection model under two condition. An simple engineering application is proposed to show its application. Finally, section 5 summarizes the main conclusions and future works.

2. Methods

2.1. Basic element theory

Extenics is a new subject founded by Chinese scholars which solves contradictory problems. It uses formal models to study the possibility of things extending and the rules and methods of development and innovation [8].

The basic-element theory in Extenics divides things into three parts: matter, affair and relation. By establishing matter-element, affair-element and relation-element, the general characteristics of matters, the interaction between affairs (i.e., an action or a behavior) and the mutual relations between affairs are described respectively [9]. The three elements of a basic-element are represented by object (O), characteristic (C), and value (V). The general expression of basic-element is denoted as:

$$B = (O, C, V) = \begin{bmatrix} \text{Object}, & c_1, & v_1 \\ & c_2 & v_2 \\ & \dots & \dots \\ & c_n & v_n \end{bmatrix} \quad (1)$$

In Formula (1), B is the name of matter-element M , affair-element A and relation-element R . O is the research object, which specifically corresponding to matter O_m , affair O_a and relation O_r . The $C = \{c_1, c_2, \dots, c_n\}$ is a collection containing various characteristics of research object. The $V = \{v_1, v_2, \dots, v_n\}$ is the characteristic value of the corresponding C_i . The establishment of matter-element, affair-element and relation-element are the process of modeling general problems. In this process, problems are recognized from different angles through the classification of basic-elements systematically [10], and the understanding of the essence of the problem is deepened in the process of mining the characteristics and value of object [9]. Another major advantage of the establishment of the base-element model is that it makes it possible to solve general problems through calculation, which is also the prominent advantage of Extenics in the rapid rise of artificial intelligence today [11].

2.2. The Extension Set and Dependent function

The Extenics usually focus on contradictory problems, transforming antithetical problems (or the discrete characteristics of an object) into consistent (or continuous) through extension transformation. Different from Cantor set and Fuzzy set which only consider quantitative relations, the extension set introduces matters and characteristics into mathematics, formalizes the process of solving contradiction problems and establishes corresponding mathematical tools to make it quantified. It discusses the research objects with certain properties and changes from the perspective of transformation. The dependent function often used to measure these changes of classification and the transformation of contradictory problems quantitatively [12].

Let the U be the domain. $P(U)$ is the power set of U , $\Omega \in P(U)$. For any element p in Ω , if there is a real number $K(p) \in (-\infty, +\infty)$ corresponding to it, then we call it is an extension set of the subset Ω over the U . Where, $y = K(p)$ is the dependent function of $\tilde{A}$, which reflects the relevance of p and y , that is, the transformation ability from p to y .

$$\tilde{A} = \{(p, y) | p \in \Omega, y = K(p)\} \quad (2)$$

An object in the domain U usually has an expected interval about the degree to which the value of a characteristic conforms to the requirements. It is called standard positive field ($X_0 = \langle a_0, b_0 \rangle$) and belongs to positive field ($X = \langle a, b \rangle, X_0 \subseteq X$) which is the interval including all possible values of x .

The concept of interval distance and place value are defined to establish the elementary dependent function. As shown in formular (3-4), the interval distance describes the distance between a point x to interval $X_0 = \langle a_0, b_0 \rangle$ in field of real numbers. And the place value describes the distance from a point x to the interval X_0 and X . The computing formulars of dependent function can be seen in formular (5).

$$\rho(x, X_0) = \left| x - \frac{a_0 + b_0}{2} \right| - \frac{b_0 - a_0}{2} \quad (3)$$

$$D(x, X_0, X) = \begin{cases} \rho(x, X) - \rho(x, X_0), & x \notin X_0 \\ -1, & x \in X_0 \end{cases} \quad (4)$$

$$K(x) = \frac{\rho(x, X_0)}{D(x, X_0, X)} \quad (5)$$

3. Matter-element modeling in AR HUD Interface

There are many interface factors that affect the visual comfort and performance of driver, such as color combination, complexity of graphic, information density, layout and display contrast, etc. Individual or conjoint analysis can be conducted among those factors by one-dimensional or multi-dimensional matter-element. Since color combination is the most significant characteristic that affects visual comfort of the AR HUD interface [13], we analyze the color combination independently in this paper.

The virtual contents presented on the AR HUD are superimposed on the physical environment through complex optical systems, making it being transparent. Therefore, ambient light, AR content color and background color will also influence the color combination of the AR HUD interface perceived by drivers. The variation of ambient light mean that accurate color and consistently high contrast ratio should be maintained to adapt to the different driving conditions in any AR HUD solutions [14].

In this section, we will conduct modeling and extended analysis on the following matter elements from the perspective of man-machine-environment: content color (machine), brightness (machine), number of applicable environments(environment) and driver performance over time (man).

3.1. Content color

The colors of visual elements such as numbers, text, graphics and symbols on the AR HUD interface are defined as the content color.

First, the basic element of content color is described as below:

$$M_1 = (\text{Content Color}, \text{RGB color model}, \text{Red}) \quad (6)$$

in which the content color is characteristic, blue is value. Furthermore, the matter element matrix can be described as formula (7):

$$M'_1 = [\text{Content Color}, \text{RGB color model}, \{\text{Red}, \text{Green}, \text{Blue}\}] \quad (7)$$

The value of content color is discontinuous and discrete. Consequently, an effective method is proposed based on the Extenics to settle the limitation.

3.2. Brightness

There are many kinds of color models including RGB, HSB, Lab and so on. HSB color model is more consistent with human visual perception, and is widely used in the field of computer vision. The correlation between the color components (H/S/B) is low and the independence is strong, which is quite suitable for the scientific calculations. And the sensitivity of human senses to brightness information is much higher than chroma [15]. Then, the computation function is defined in formula (8)-(12):

$$R' = R/255 \quad (8)$$

$$G' = G/255 \quad (9)$$

$$B' = B/255 \quad (10)$$

$$C_{max} = \max(R', G', B') \quad (11)$$

$$\text{Brightness} = C_{max} \quad (12)$$

According to the above conversion relationship, the brightness of white is 100% while black is 0%. Then, an extended matter-element model based on content color is established as follows.

$$M_2 = (\text{Content color}, \text{Brightness}, x\%) \quad (13)$$

$$M'_2 = [\text{Content Color}, \text{Brightness}, \{100\%, 90\%, \dots, x\%\}] \quad (14)$$

Now, the brightness is consistent and continuum.

3.3. Number of applicable environments

The display effect of the AR HUD projection perceived by human eyes is determined by the brightness of projected light and ambient light [16]. In AR HUD, the virtual image is viewed through transparent windshield, and the luminance contrast between the image and the background is a very critical factor which can affect visual performance and subjective preference [17]. Many articles have proposed that visual search performance improves with the increase of luminance contrast [18-20]. H.C. Weston's results show that the effect of luminance contrast in visual performance has always been critical, and no matter how high the brightness, poor contrast performance can never be compared with good contrast performance [21]. Ya-Hsien Ko also found that legibility was significantly impacted by luminance contrast, with low contrast sensitivity making it difficult to perceive faces, objects, or road signs [22].

Previous researches suggested that the visual display terminals should have a luminance contrast of at least 1:3 [23-25]. For example, we suppose the brightness units are ranged from 0 to 100. When the content color has a brightness unit of 30, the environment should have a brightness unit within 0-10 and 90-100 (3 times unit difference) to ensure the readability in ambient light. We define the number of applicable environments as the number of ambient brightness unit that meet the display contrast requirements for a given content color brightness. Thus, the number of applicable environments is 22. In the same way, if the brightness unit of content color is 90, the number of applicable environments is 31(0-30).

Within a certain range, the AR content with high brightness can be applied to more environments, which means that drivers are more sensitive to the information presented by the AR HUD interface and have better driving performance in a short time. On the contrary, AR content with low brightness can be applied to fewer environments, resulting in lower driver sensitivity and poor driving performance in a short time. Therefore, if only the number of applicable environments is considered, content color with high brightness can achieve better driving performance in a short while than those with low brightness.

Then, the number of applicable environments is expanded to quantify the brightness, and its matter-element model is constructed as M_3 and M'_3 .

$$M_3 = (\text{Environment}, \text{Number of applicable environments}, n) \quad (15)$$

$$M'_3 = \begin{pmatrix} \text{Environment}, & \text{Number of applicable environments}, & 1 \\ \text{Environment}, & \text{Number of applicable environments}, & 2 \\ \dots & \dots & \dots \\ \text{Environment}, & \text{Number of applicable environments}, & n \end{pmatrix} \quad (16)$$

3.4. Driver performance over time

However, there are also many studies have suggested that working with high screen brightness for long periods of time may cause visual fatigue. Simone et al. studied the interaction effect of screen brightness and ambient illuminance on visual fatigue and arousal during prolonged digital reading (1 hour). The results showed that both visual fatigue and arousal increased simultaneously under high screen brightness [26]. While the rise of arousal has a good connotation for improving reading performance, an upsurge in visual fatigue might be a bad new for our eyes. Wang et al. found that there was a negative correlation between the roadside landscape color and drivers' EEG δ wave components, that is, the higher the brightness of the landscape color, the more effective it was to avoid visual fatigue [27]. Joseph L. Gabbard and his colleagues indicated that repeatedly performing context switching and focal distance switching between a real and AR environment can cause the visual fatigue steadily increase and significantly reduce performance [28].

According to the literature review above, we conducted a qualitative analysis on brightness and visual fatigue degree. That is, the greater the luminance contrast, the stronger the stimulation to human eyes, prolonged viewing exacerbates visual fatigue. While the lower luminance contrast could reduce the sensitivity of the eye to flicker, leading to better fatigue resistance. Thus, if only driver performance over time (DOT) is considered, content colors with low brightness achieve better over long periods of time than the high one.

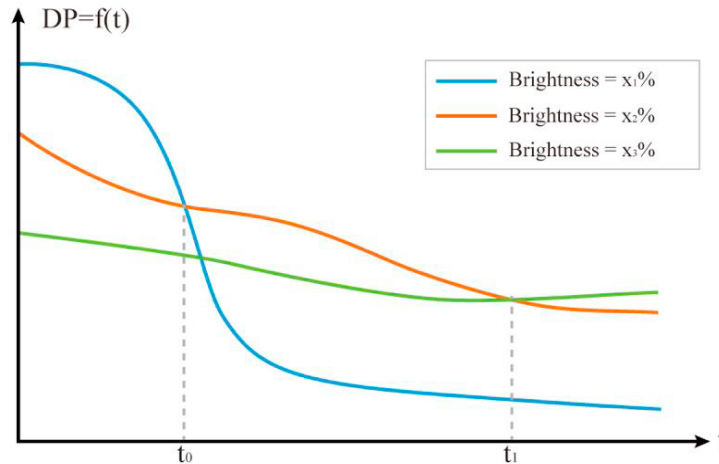


Fig. 2. Driver performance over time at different brightness ($x_1 > x_2 > x_3$).

As shown in Fig.2, DP at different brightness will decline at different speed when eyes viewing on the AR HUD interface over time ($x_1, x_2, x_3 \in [50, 100], x_1 > x_2 > x_3$). The Formula (17) describes the calculation function for DOT, which is the performance integral over time, approximated as a curve T (Fig. 2).

$$DOT = \int_0^T f(t) dt \quad (17)$$

The matter-element can be described as M_5 :

$$M_5 = (\text{Driver}, \text{Performance over time}, t) \quad (18)$$

The matter-element array of brightness on DOT can be described as M'_5 :

$$M'_5 = \begin{pmatrix} \text{Driver, Performance over time,} & 1 \\ \text{Driver, Performance over time,} & 2 \\ \dots & \dots \\ \text{Driver, Performance over time,} & t \end{pmatrix} \quad (19)$$

To sum up, the contradiction in color design of AR HUD interface is that if content colors with high brightness are selected, the visual sensitivity of drivers is improved, but only perform well in a short time. If the content color with low brightness is selected, the visual fatigue degree of long-term driving is reduced, but the sensitivity is sacrificed.

3.5. Mapping relation and matter-element transformation

The process of matter-element transformation based on Extension analysis is shown in Fig.3. Firstly, the correlation analysis of content color is carried out, and the related characteristic-color model-is mined. Then, according to the divergence analysis method of ‘one characteristic has multiple values’, the HSB model is diverged from the RGB model, and the brightness component which can quantitatively describe the content color is obtained. Since the optimal solution of the brightness value of the content color is not always known, the driver and environment characteristics that can quantitatively describe the brightness of the AR HUD interface are further expanded from the perspective of man-machine-environment. Among these characteristics, content color and brightness can be converted into each other, brightness and NAE, DOT can be converted into each other. It means that discrete and opposing variable are transformed into continuous and consistent ones.

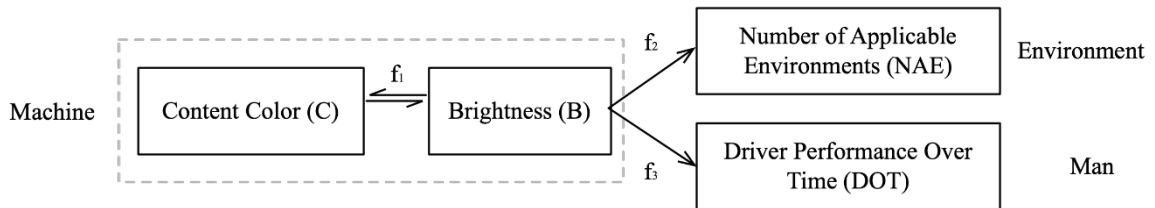


Fig. 3. The process of matter-element transformation.

The relationships among brightness, NAE and DOT are shown in Fig. 4. The NAE at different brightness are summarized as curve N by satisfying luminance contrast requirements, and the DOT are approximately draw as curve T by time integration.

- When brightness approach 0% or 100%, the NAE is best, that is, both lower and higher brightness can theoretically be applicable in more environments.
- When the brightness close to a%, the NAE is the worst, due to the low luminance contrast between the AR interface and the environment.
- When brightness approach to b%, the DOT is best, which means only the appropriate brightness can ensure the long driving performance.

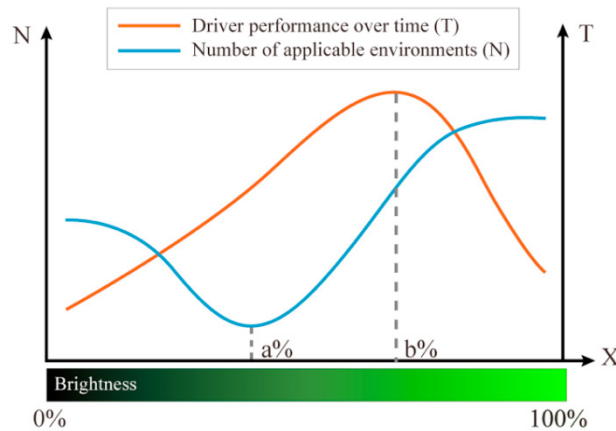


Fig. 4. The relationship between brightness and NAE, DOT.

Now, the dilemma problem in AR HUD interface design has been transformed into finding a proper brightness that balance the NAE and DOT, so that the driver comprehensive visual performance to achieve the best. In the next section, we will establish the color selection model, helping engineers and designers to design graphics or character colors more scientifically and systematically.

4. Color selection models for AR HUD interface

For the optimal solution of content color, the distance function from the selected matter-element to optimal value is constructed in this chapter.

In Fig. 5, the given standard positive fields of NAE and DOT are N_0 and T_0 respectively (N_0 and T_0 are determined by the engineering). The positive field of the brightness is $X = \langle 0\%, 100\% \rangle$, where the expected interval that meet the requirements of NAE and DOT is $X_0 = \langle p_0, q_0 \rangle$ and x is the selected brightness value.

When the optimal value of brightness has been given, it is only required to verify whether x is within the interval X_0 and how relevant is it to the best value. The calculation model based on the brightness is established in the first section of this chapter. If the optimal value is uncertain, the x needs to be evaluated by NAE and DOT. The calculation model based on the NAE and DOT is established in the second section of this chapter.

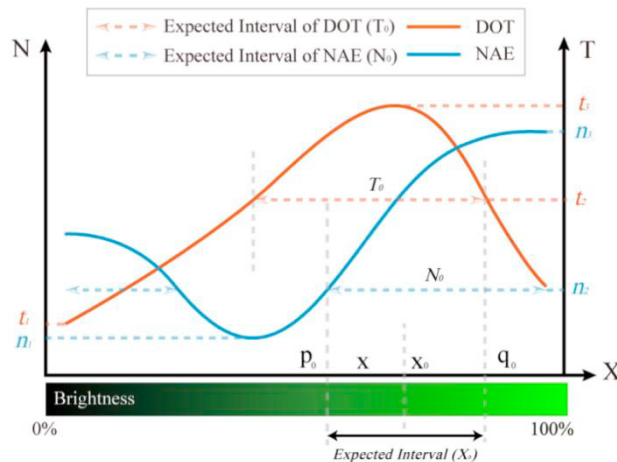


Fig.5. Color selection model for AR HUD interface.

4.1. Color select model based on the brightness

The x_0 is defined as the optimal value in the X_0 . When the content color is selected, $BL = x$ will be mapped in X_0 , the distance from x to X_0 can be described as $\rho = (x, X_0)$, the relevancy between x and x_0 is $K(x)$. The calculation formula of $K(x)$ is described as follows (20-23).

$$\rho(x, X_0) = \left| x - \frac{1}{2}(p_0 + q_0) \right| - \frac{1}{2}(q_0 - p_0) \quad (20)$$

$$\begin{aligned} \rho(x, X) &= \left| x - \frac{1}{2}(0\% + 100\%) \right| - \frac{1}{2}(100\% - 0\%) \\ &= \begin{cases} x - 100\%, & x \geq 50\% \\ x, & x < 50\% \end{cases} \end{aligned} \quad (21)$$

$$D(x, X_0, X) = \begin{cases} \rho(x, X) - \rho(x, X_0), & x \notin X_0 \\ -1, & x \in X_0 \end{cases} \quad (22)$$

$$\begin{aligned} K(x) &= \frac{\rho(x, X_0)}{D(x, X_0, X)} = \frac{\left| x - \frac{1}{2}(p_0 + q_0) \right| - \frac{1}{2}(q_0 - p_0)}{D(x, X_0, X)} \\ &= \begin{cases} \frac{\left| x - \frac{1}{2}(p_0 + q_0) \right| - \frac{1}{2}(q_0 - p_0)}{-1}, & x \in X_0 \\ \frac{\left| x - \frac{1}{2}(p_0 + q_0) \right| - \frac{1}{2}(q_0 - p_0)}{\rho(x, X) - \rho(x, X_0)}, & x \notin X_0 \end{cases} \\ &= \begin{cases} \frac{\left| x - \frac{1}{2}(p_0 + q_0) \right| - \frac{1}{2}(q_0 - p_0)}{-1}, & x \in X_0 \\ \frac{\left| x - \frac{1}{2}(p_0 + q_0) \right| - \frac{1}{2}(q_0 - p_0)}{\begin{cases} x - 100\%, & x \geq 50\% \\ x, & x < 50\% \end{cases} - \left(\left| x - \frac{1}{2}(p_0 + q_0) \right| - \frac{1}{2}(q_0 - p_0) \right)}, & x \notin X_0 \end{cases} \end{aligned} \quad (23)$$

If $K(x) < 0$, then x is not in X_0 , and the selected color does not satisfy the required NAE and DOT. If $K(x) > 0$, it means that x is in the interval of X_0 , and the closer $K(x)$ is to 0, the better the selection. The optimal solution for $K(x)$ is $n - x_0$ (if $x_0 > \frac{1}{2}(p_0 + q_0)$) or $x_0 - m$ (if $x_0 < \frac{1}{2}(p_0 + q_0)$).

The formula indicates that the correlation between mapped matter-element m and the optimal value of mapped matter-element m_0 when $BL = x$.

4.2. Color select model based on the NAE and DOT

If x_0 is unknown, then x should be evaluated by the best values of NAE and DOT respectively. The calculation model in this case can be formulated as:

$$K(m)|_{BL=x} = \frac{\rho(m, M_0)}{D(m, M_0, M)}|_{BL=x} \quad (24)$$

The formula indicates that the correlation between mapped matter-element m and the optimal value of mapped matter-element m_0 when $BL = x$.

The optimal value of NAE is defined as n_0 , the whole interval of NML is $N = [n_1, n_3]$, and the expected interval is $N_0 = [n_2, n_3]$. The distance between n and N_0 is $\rho(n, n_3) = |n - n_3|$. The correlation between brightness and the optimal value of its mapped-element NAE is described in (25-26).

$$D(n, N_0, N) = \begin{cases} \rho(n, N) - \rho(n, N_0), & n \notin N_0 \\ -1, & n \in N_0 \end{cases} \quad (25)$$

$$K(n)|_{BL=x} = \frac{\rho(n, n_3)}{D(n, N_0, N)}|_{BL=x} \quad (26)$$

The optimal value of DOT is defined as t_0 , the whole interval of DOT is $T = [t_1, t_3]$, and the expected interval is $T_0 = [t_2, t_3]$. The distance between t and t_3 is $\rho(t, t_3) = |t - t_3|$. The correlation between brightness and the optimal value of its mapped-element DOT is described in (27-28).

$$D(t, T_0, T) = \begin{cases} \rho(t, T) - \rho(t, T_0), & t \notin T_0 \\ -1, & t \in T_0 \end{cases} \quad (27)$$

$$K(t)|_{BL=x} = \frac{\rho(t, t_3)}{D(t, T_0, T)}|_{BL=x} \quad (28)$$

When $K(n)$ and $K(t)$ are both closer to 0, the solution is better. The calculation formula of $K(x)$ can be defined as follows:

$$\begin{aligned} K(n)|_{BL=x} &= \frac{\rho(n, n_3)}{D(n, N_0, N)}|_{BL=x} = \frac{|n - n_3|}{D(n, N_0, N)}|_{BL=x} \\ &= \begin{cases} \frac{|n - n_3|}{-1}|_{BL=x}, & n \in N_0 \\ \frac{|n - n_3|}{\rho(n, N) - \rho(n, N_0)}, & n \notin N_0 \end{cases} \\ &= \begin{cases} \frac{|n - n_3|}{-1}|_{BL=x}, & n \in N_0 \\ \frac{|n - n_3|}{\left(\left|\frac{n_1 + n_3}{2}\right| - \left|\frac{n_3 - n_1}{2}\right|\right) - \left(\left|\frac{n_3 + n_2}{2}\right| - \left|\frac{n_3 - n_2}{2}\right|\right)}, & n \notin N_0 \end{cases} \end{aligned} \quad (29)$$

$$\begin{aligned} K(t)|_{BL=x} &= \frac{\rho(t, t_3)}{D(t, T_0, T)}|_{BL=x} = \frac{|t - t_3|}{D(t, T_0, T)}|_{BL=x} \\ &= \begin{cases} \frac{|t - t_3|}{-1}|_{BL=x}, & t \in T_0 \\ \frac{|t - t_3|}{\rho(t, T) - \rho(t, T_0)}, & t \notin T_0 \end{cases} \\ &= \begin{cases} \frac{|t - t_3|}{-1}|_{BL=x}, & t \in T_0 \\ \frac{|t - t_3|}{\left(\left|\frac{t_1 + t_3}{2}\right| - \left|\frac{t_3 - t_1}{2}\right|\right) - \left(\left|\frac{t_3 + t_2}{2}\right| - \left|\frac{t_3 - t_2}{2}\right|\right)}, & t \notin T_0 \end{cases} \end{aligned} \quad (30)$$

$$K(x) = K(n)|_{BL=x} + K(t)|_{BL=x} \quad (31)$$

4.3. Weighted color select model

As shown in Fig. 6, the driver performance over time varies with the brightness of the content color.

The information on $x_1\%$ brightness has a higher sensitivity at the beginning, and the driver performance is better than other brightness, but decreased after t_0 due to increased visual fatigue.

The $x_2\%$ brightness is lower than $x_1\%$, so the driver performance is not as good as it at the beginning, but after t_0 , the drop rate in driver performance due to fatigue is lower than $x_1\%$, resulting in better driving performance.

The driver performance on $x_3\%$ brightness changed slowest with the lapse of time, showing the best fatigue defense.

In the AR HUD interface design, the color should be selected according to the attributes of content information, such as viewing time, priority, state variation, etc. For example, the content with low priority and long viewing time can choose content color with lower brightness. And for content with high priority and short viewing time, a content color with higher brightness can be selected. Consequently, different weights can be assigned to the indicators affecting color selection to build a weighted color selection model. In this section, we define the weight of NAE ($K(n)$) as $w\%$, and the weight of DOT ($K(t)$) as $1 - w\%$. The preliminary model can be optimized as formula (32). Similarly, the closer the value of $K'(x)$ is to 0, the smaller distance from the selected value x to the optimal solution.

$$K'(x) = w\% * K(n)|_{BL=x} + (1 - w\%) * K(t)|_{BL=x} \quad (32)$$

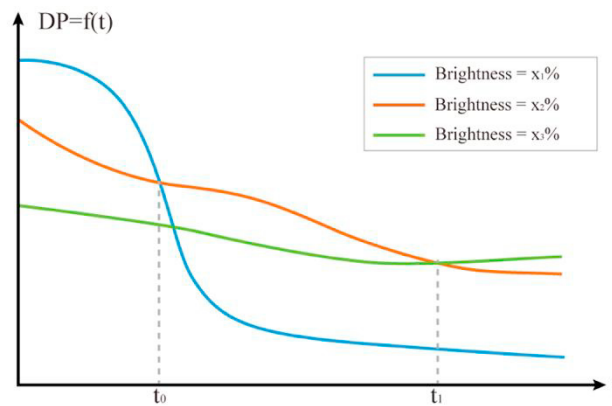


Fig. 6. Performance over time (DP: Driver performance).

4.4. Engineering application

Considering that different luminance contrasts have positive and negative effects on driver performance, the application of this model in design engineering is discussed in this section. For the content with low information priority and long viewing time, anti-fatigue should be considered first, which is reflected in the model as increasing the weight of DOT and reflected in the color as low brightness. If choose $w\% \approx 20\%$, $BL \approx 65$, then $K(n) \approx 2$, $K(t) \approx 0$, $K'(x) = 20\% * K(n) + 80\% * K(t) = 0.4$.

On the contrary, for the content with high information priority and short viewing time, reaction speed should be considered first, which is reflected in the model as increasing the weight of NAE, and reflected in the color as high brightness. If choose $w\% \approx 100\%$, $BL \approx 100$, then $K(n) \approx 0$, $K(t) \approx 4$, $K'(x) = 1 * K(n) + 0 * K(t) = 0$.

According to this model, optimization suggestions are proposed for the design application of AR HUD in the intelligent cockpit. The AR HUD interface content mainly consists of basic driving information, safety information and multimedia entertainment information. Basic driving information (including average speed, remaining fuel, map navigation, etc.) will be displayed continuously during driving, and a brightness of 65% is selected as the content color ($w\% \approx 20\%$). Multimedia entertainment information (including weather, music, video, communication, etc.) is only viewed during operation, so this type of information has a short viewing time and low priority, and a brightness of 75% is selected as the content color ($w\% \approx 30\%$). As for the safety information (including obstacle information, lane track information and traffic sign information), 100% brightness ($w\% \approx 0\%$) is selected as the content color because

of its short viewing time and high demand for reaction speed. In addition, a certain information type can be further subdivided, for example, pedestrian and obstacle information are high weight information, and lane trajectory and traffic sign information are secondary weight information, so the brightness of obstacle information should be larger than traffic sign information.

5. Conclusions

The AR HUD will provide advanced information on security and entertainment in intelligent vehicles in the near future. By visualizing lanes, cars, pedestrians, and predicting the expected path of cars, it can build public trust in autonomous driving. Based on the matter element analysis and dependent function, the paper transforms the discrete problem into continuous one, and provides a solution about the content color selection in AR HUD interface design. The mapping relationship between the content color and brightness of AR HUD is analyzed qualitatively, and a quantitative weighted color selection model is established to reflect the tendency towards selected elements to the optimal interval by constructing dependent functions. Finally, a simple evaluation method and an application case are given.

In future studies, we will conduct series human factors experiments by driving simulators, and collect more engineering data to conduct a more accurate and specific quantitative analysis of the relationship between brightness and DOT and NML. Through continuous modification of matter-element relationship, change curve, optimal interval and weight value, the final calculation result is more scientific and reasonable, and provides effective support for the optimization of AR HUD interface design in intelligent vehicle.

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